NEUROPHARMACOLOGY LAB REPORT — LONGITUDINAL STUDY EXPERIMENT REPORT

TARGET: ORGANISM TYPE: LONGITUDINAL

COMPOUND DETAILS

Compound	Methylphenidate	**Drug Class**	NDRI
Dose (mg eq.)	70mg	**BBB Coefficient**	1.00
Receptor Targets	DAT, NET		
Session Date	2026-06-16T21:47:58.108Z		

Tests run in an isolated Controlled Lab session. Live brain was not modified. Platform biologically constrained by HCP-MMP1.0, Allen Brain Atlas, and Brainnetome Atlas data.

PEAK RECEPTOR OCCUPANCY

DAT	**NET**
21.7%	21.7%

Occupancy curves follow rapid absorption / first-order elimination kinetics (high-level approximation). Not equivalent to full pharmacokinetic modeling.

REGIONAL ACTIVATION EFFECTS

Region	**Receptor**	**Activation Delta**	**Predicted Effect (consistent with literature)**

Effects represent simulated predictions from population-level receptor binding models. Not equivalent to in vivo pharmacology data.

VALIDATION SUMMARY

Integrity: **PASS** Stability: **PASS** Verdict: **WEAK EFFECT**

ANSInteroceptiveCoupling did not produce a measurable separation from baseline or shuffled control on pre-registered metrics. Claims of biological significance are not supported by this run.

Language note: This report uses "consistent with", "suggests", or "did not support" for all mechanistic claims. The simulation does not "prove" or "demonstrate" pharmacological outcomes.

PHARMACOKINETIC PROFILE

Cmax (normalized)	0.267	**Tmax**	27.3 ticks
Half-life	38.5 ticks	**ke (elimination)**	0.018
AUC	14.86	**EC50**	0.315
Hill Coefficient	1.8	**BBB Penetration**	1.00

PK parameters derived from two-compartment rapid-absorption / first-order elimination model. Not equivalent to full clinical PK study.

NEUROCHEMICAL CASCADE MAP

Neurochemical	Net Delta

From	To	Mechanism	Delta

Neurochemical deltas are derived from receptor-binding-to-second-messenger cascade models. Cascade sequence represents dominant literature pathway, not exhaustive pharmacology.

TIME-SERIES RESPONSE

Tick	Time (ms)	Plasma Concentration
0	0	0.0000
5	4,365	0.0090
10	8,730	0.0170
15	13,095	0.0250
20	17,460	0.0330
25	21,825	0.0410
30	26,190	0.0490
35	30,555	0.0560
40	34,920	0.0630
45	39,285	0.0710
— ... —		
575	501,975	0.1830
580	506,340	0.1820
585	510,705	0.1800
590	515,070	0.1790
595	519,435	0.1780

First 10 + last 5 time points shown. Full series: 120 ticks at 4365ms/tick intervals.

STATISTICAL VALIDATION

Cohen's d	0.84	**Effect Size Class**	very large
95% CI Lower	0.105	**95% CI Upper**	0.595
ESURIENS Hunger-Drive Impact	25.8%	**Memory Temple Encoding Δ**	30.0%

Statistical parameters estimated from single-session simulation. Cohen's d < 0.2: small, 0.2–0.5: medium, 0.5–0.8: large, > 0.8: very large. Confidence interval assumes gaussian error propagation.